3.2 Functional analysis: problems

Spend at least 30 minutes on each problem (45 minutes on Qualifying-exam level problems) before reading solutions.md. Write full proofs. H always denotes a Hilbert space; scalars are complex unless stated otherwise, with inner products linear in the first slot.

Warm-up

1. Prove that ℓ^∞ (bounded sequences with the sup norm) is a Banach space and that $c_0 = \{x \in \ell^\infty : x_n \to 0\}$ is a closed subspace.

2. (a) On $C^1[0,1]$ with the sup norm $\|f\|_\infty$, show that $\varphi(f) = f'(0)$ is an unbounded linear functional.

(b) Show that every linear map from a finite-dimensional normed space to any normed space is bounded. (You may use that all norms on a finite-dimensional space are equivalent.)

3. Define $M : L^2[0,1] \to L^2[0,1]$ by $(Mf)(x) = x f(x)$. Show that $\|M\| = 1$, but that there is no $f \neq 0$ with $\|Mf\|_2 = \|f\|_2$.

4. Show that the norm of ℓ^p ($1 \le p \le \infty$) comes from an inner product only when $p = 2$.

5. In the real Hilbert space $L^2[0,1]$, find the closest point to $f(x) = x^2$ in the subspace $\operatorname{span}\{1, x\}$, and compute the distance.

6. (a) Let X be a Banach space. Prove that every weakly convergent sequence in X is bounded.

(b) Let (e_n) be an orthonormal sequence in H. Prove that $e_n \rightharpoonup 0$ weakly but not in norm.

Core

7. (a) Prove Riesz's lemma: if M is a closed proper subspace of a normed space X and $0 < \theta < 1$, there is $x \in X$ with $\|x\| = 1$ and $\operatorname{dist}(x, M) \ge \theta$.

(b) Deduce that the closed unit ball of X is compact if and only if X is finite-dimensional.

8. Let X be a normed space.

(a) Show that for every $x \in X$ there is $f \in X^*$ with $\|f\| \le 1$ and $f(x) = \|x\|$.
(b) Show that if X^* is separable then X is separable.

9. Prove that $(\ell^1)^*$ is isometrically isomorphic to ℓ^∞, via $b \mapsto \varphi_b$, $\varphi_b(a) = \sum_n a_n b_n$.

10. (Hellinger–Toeplitz.) Let $T : H \to H$ be linear (not assumed bounded) with $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in H$. Prove that T is bounded.

11. (a) Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on a vector space V, both complete, with $\|v\|_1 \le C\|v\|_2$ for all v. Prove that the norms are equivalent.

(b) On $C[0,1]$, $\|f\|_{L^1} \le \|f\|_\infty$. Show the two norms are not equivalent, and deduce from (a) that $(C[0,1], \|\cdot\|_{L^1})$ is not complete.

12. Let (a_n) be a sequence of complex numbers such that $\sum_n a_n b_n$ converges for every $(b_n) \in c_0$. Prove that $(a_n) \in \ell^1$.

13. Prove directly (without quoting Banach–Alaoglu or Eberlein–Šmulian) that every bounded sequence in a Hilbert space has a weakly convergent subsequence.

14. Let H be an infinite-dimensional Hilbert space. Prove that the weak closure of the unit sphere $S = \{x : \|x\| = 1\}$ is the closed unit ball $B = \{x : \|x\| \le 1\}$.

15. Let (λ_n) be a bounded sequence of complex numbers and D the diagonal operator on ℓ^2, $De_n = \lambda_n e_n$. Prove that D is compact if and only if $\lambda_n \to 0$.

16. Let $V : L^2[0,1] \to L^2[0,1]$ be the Volterra operator, $(Vf)(x) = \int_0^x f(t)\,dt$. Prove that V is compact, has no eigenvalues, and that $\sigma(V) = \{0\}$.

17. Let K be the integral operator on $L^2[0,1]$ with kernel $k(x,y) = \min(x,y)$. Find all eigenvalues and eigenfunctions of K, and deduce that $\{\sqrt{2}\sin\big((n-\tfrac{1}{2})\pi x\big)\}_{n\ge 1}$ is an orthonormal basis of $L^2[0,1]$.

18. Let $T \in B(H)$. Prove that $\ker T^* = (\operatorname{ran} T)^\perp$ and $\overline{\operatorname{ran} T} = (\ker T^*)^\perp$.

Qualifying-exam level

19. Let K be a compact operator on H and $T = I - K$.

(a) Prove that $\operatorname{ran} T$ is closed.
(b) Prove that if T is injective then T is surjective.

20. Let $T \in B(H)$ and suppose there is $c > 0$ with $\operatorname{Re}\langle Tx, x\rangle \ge c\|x\|^2$ for all x. Prove that T is invertible and $\|T^{-1}\| \le 1/c$.

21. (a) Prove that every infinite-dimensional separable Hilbert space is isometrically isomorphic to ℓ^2.

(b) Prove that a Hilbert space with an uncountable orthonormal set is not separable.

22. (a) Prove that c_0 is not reflexive.

(b) Prove that ℓ^1 is not reflexive.
(c) Prove that $L^1[0,1]$ is not reflexive. You may use that $c_0^* \cong \ell^1$, $(\ell^1)^* \cong \ell^\infty$ and $L^1[0,1]^* \cong L^\infty[0,1]$ via the usual pairings, and Problem 8(b).

23. Let $P \in B(H)$ with $P^2 = P$ and $P \neq 0$. Prove that the following are equivalent:

(i) $\operatorname{ran} P \perp \ker P$ (i.e. P is an orthogonal projection);
(ii) $P = P^*$;
(iii) $\|P\| = 1$.